

How Cookie Consent Interfaces Changed

A controlled capture method plus six dated company trajectories from primary sources

Qianyi (Alfred) Chen | Mentor: Dr. Jagdip Singh

RQ1	component rubric
RQ2	dated trajectories
RESULT	5 improved 1 regressed

direction is separated from causal evidence strength

THESIS Concrete interactions improved under regulatory pressure, but consent quality remained component-specific and reversible.

01 QUESTION + MEASURE

How did consent interfaces change over time, and what evidence can explain the direction?

RQ1 codes user choice by component. RQ2 compares dated states while keeping direction separate from cause.

RQ1: the component ruler

Path availability

Can users reach Reject and Customize?

Path effort

How much interaction does each path require?

Transparency

What is disclosed, and how clearly?

Choice + effectiveness

Are options balanced, and does refusal stop tracking?

Directional coding

IMPROVED	one or more dimensions improve; none regress
REGRESSED	one or more regress; none improve
MIXED	dimensions move in both directions
STABLE	no meaningful component change
INSUFFICIENT	context, evidence, or scoring is not comparable

Claim guardrail

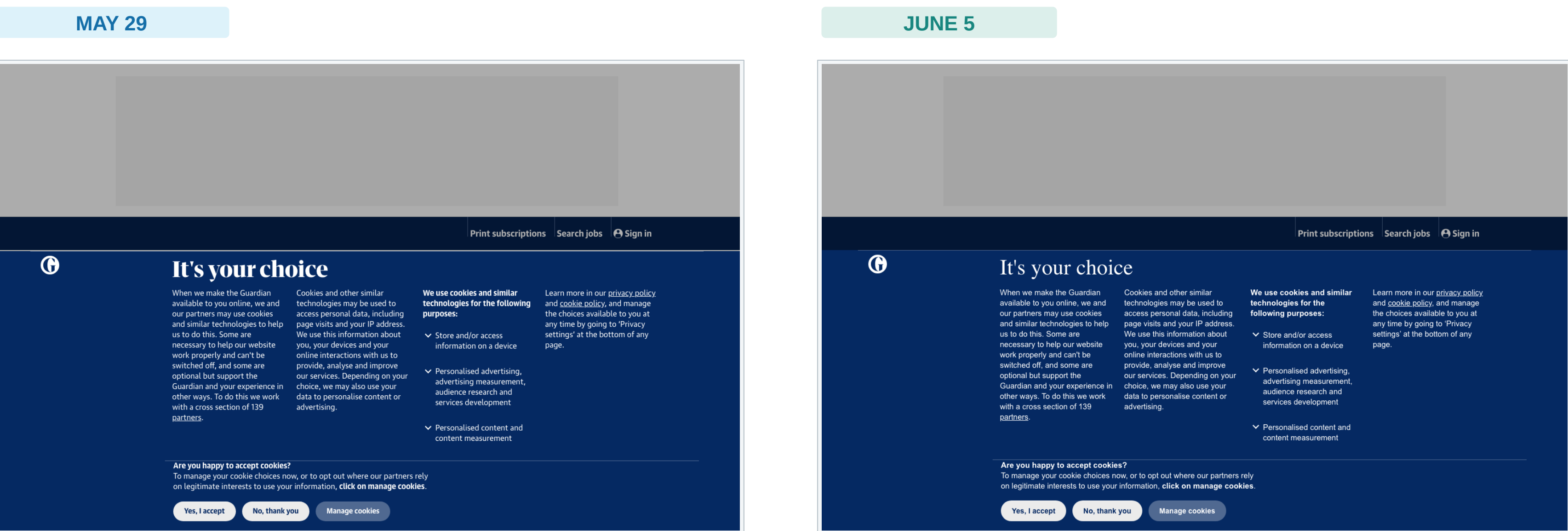
Every direction needs dated before/after evidence. Every causal claim receives a separate strength label.

02 TWO EVIDENCE LANES

Controlled local capture + source-complete retrospective cases

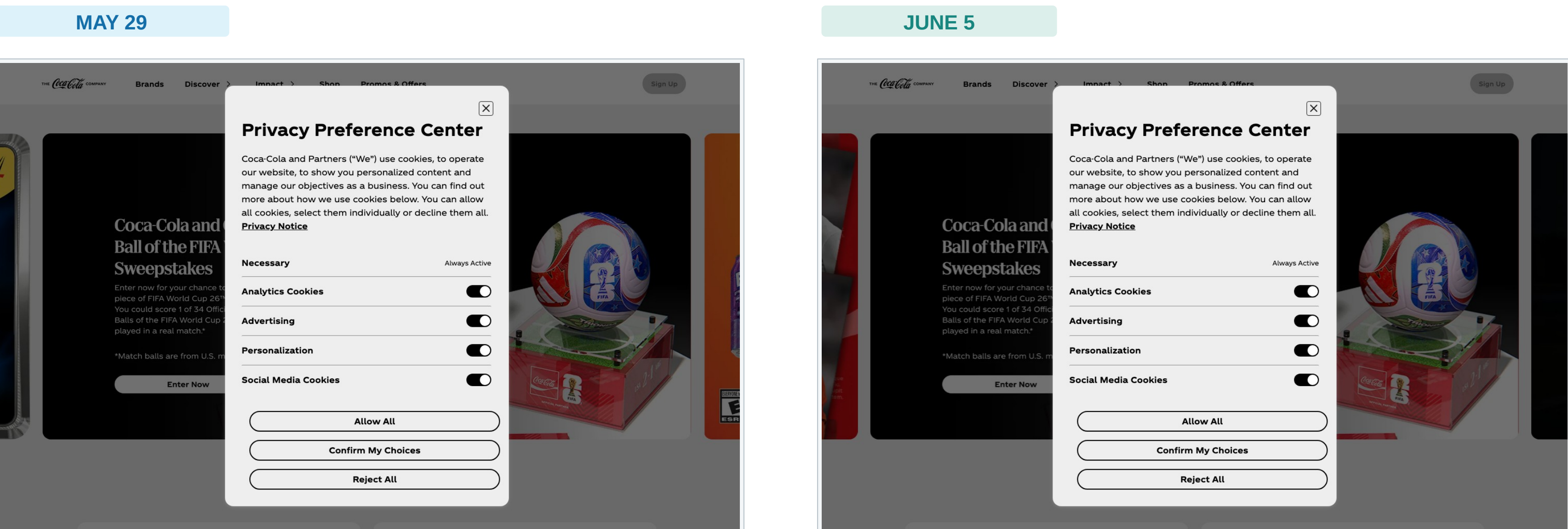


Controlled pilot 1: The Guardian



Local lane: the same visible choice structure remains insufficient for a direction claim

Controlled pilot 2: Coca-Cola



Local lane validates the comparison method. Historical direction comes from six dated primary-source cases.

03 CASE SERIES RESULT + DISCUSSION

6 companies

5 improved

1 regressed

RETROSPECTIVE RESULT

Across six purposively selected dated trajectories, five improved in at least one audited component; Vanity Fair regressed in functional refusal. This is not a prevalence estimate.

Three first-layer cases

Google, Facebook, and TikTok moved from acceptance-favoring rejection effort to one-click or equivalent first-layer refusal. TikTok still had deficient purpose information.

What changed, and what explains it?

- Google: >=5 -> 1 reject action; direct attribution
- Facebook: equivalent reject; order verified
- TikTok: first-layer reject; investigation-period change
- Orange: withdrawal fixed; order verified
- SHEIN: refusal/withdrawal fixed during proceedings
- Vanity Fair: later functional failure; cause unknown

Discussion prompts

- 1 Visible parity or genuine user autonomy?
- 2 Publisher, CMP, or both when refusal fails?
- 3 One-time certification or recurring re-audit?
- 4 Which component changes first under enforcement?
- 5 Can controlled captures reproduce these cases?

DISCUSSION FINDING

Regulators appear most effective where they can specify and verify a concrete interaction.

CONCLUSION

Consent interfaces improved where concrete interactions were specified and checked, but improvement was partial and reversible.

Methods and evidence

CONCEPTS.md | retrospective cases.csv | source registry.csv | local directional review